Forest Research

NCEA: Tree Species Mapping Methodology

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NCEA | Natural Capital and Ecosystem Assessment

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1 Introduction

This report outlines the approach and decisions made to produce a distribution map of common tree species in England using optical satellite remote sensing data. We trained a machine-learning classifier with the powerful XGBoost algorithm to discriminate between broadleaved and conifer species. The classifier was trained using features derived from time-series Sentinel-2 surface reflectance (SR) imagery and labelled training data obtained from the sub-compartment database.

Section 2 describes the remote sensing data used in the classifier and demonstrates the effectiveness of Sentinel-2 bands for the classification of trees at the species level. In Section 3, we address modelling challenges such as class imbalances and missing data and outline the model biases and assumptions. Our test data results are given in Section 4, where we also describe the products derived from the model predictions. Section 5 provides an overview of our approach to validating our results.

Data availability

Please contact the corresponding author for access to the data and code used in this work

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2 Data

2.1 Data Exploration

The mean Sentinel-2 SR measurements in the late spring and autumn of 2019 for all 10-20m bands for several pine (Figure 1) and oak (Figure 2) species found in England are plotted below. Pixel data is sampled from mono-species stands, as labelled in the Sub-Compartment Database (SCDB). Encouragingly, this data shows distinct spectral differences between species, particularly in the regimes of the vegetation red-edge and SWIR, and so should be applicable for the purpose of species classification.

Some species are more spectrally distinct in certain seasons, and the inter-seasonal spectral change vary between species (figure 1 and 2). Further, as tree phenology varies from year to year in response to the changing climate, we include multiple years of data for species classification tasks. Median composites for late spring (1st April - 30th June) and autumn (1st September - 30 November) were created for 2019, 2020, 2021 and 2022. The decision to generate composites over 3 months may have advantages in averaging over both regional differences in plant phenology, and any artifacts in the imagery due to imperfections in cloud and shadow masking.

In Figure 3 we plot the mean 2019 S2 SR measurements for several of the more common larch, spruce, pine and fir species, colour-coded by genus. This hints at substantial spectral variation for species within a genus, particularly for pine and spruce, and so motivates constructing a classifier at the species, rather than genus level.

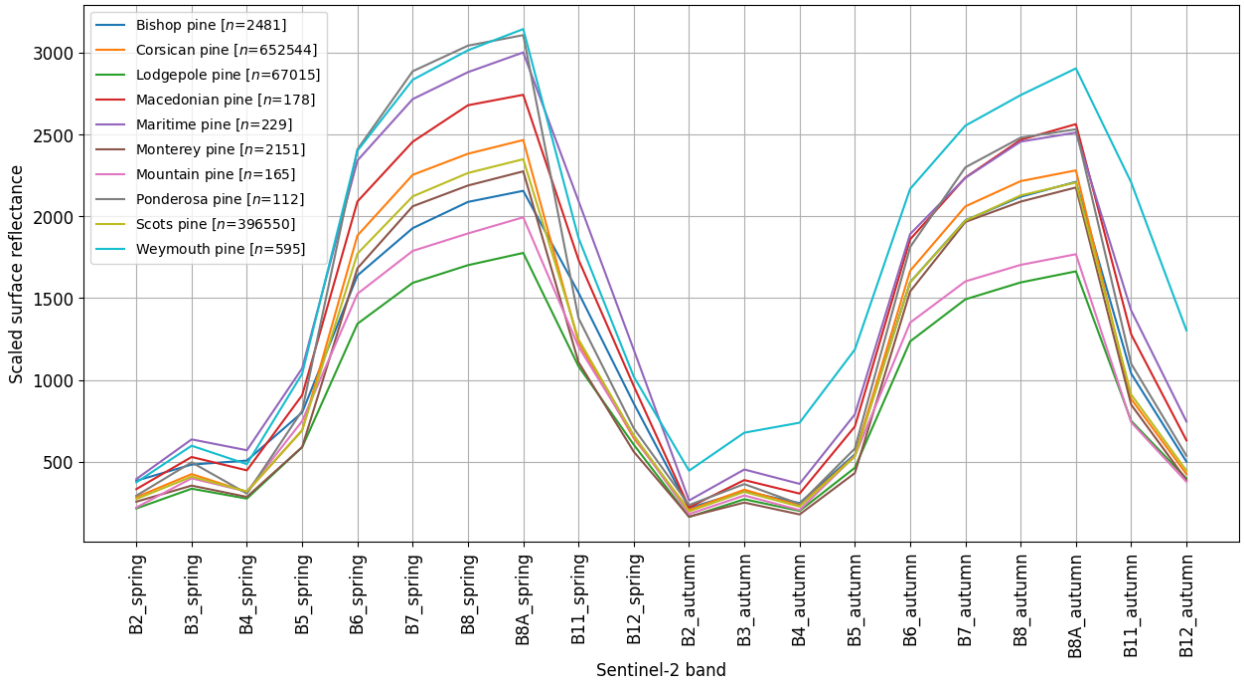

Figure 1: Mean S2 band SR measurements in the spring and autumn of 2019 for several pine species. The number of *n* samples used to compute each mean band SR measurement is shown in the legend.

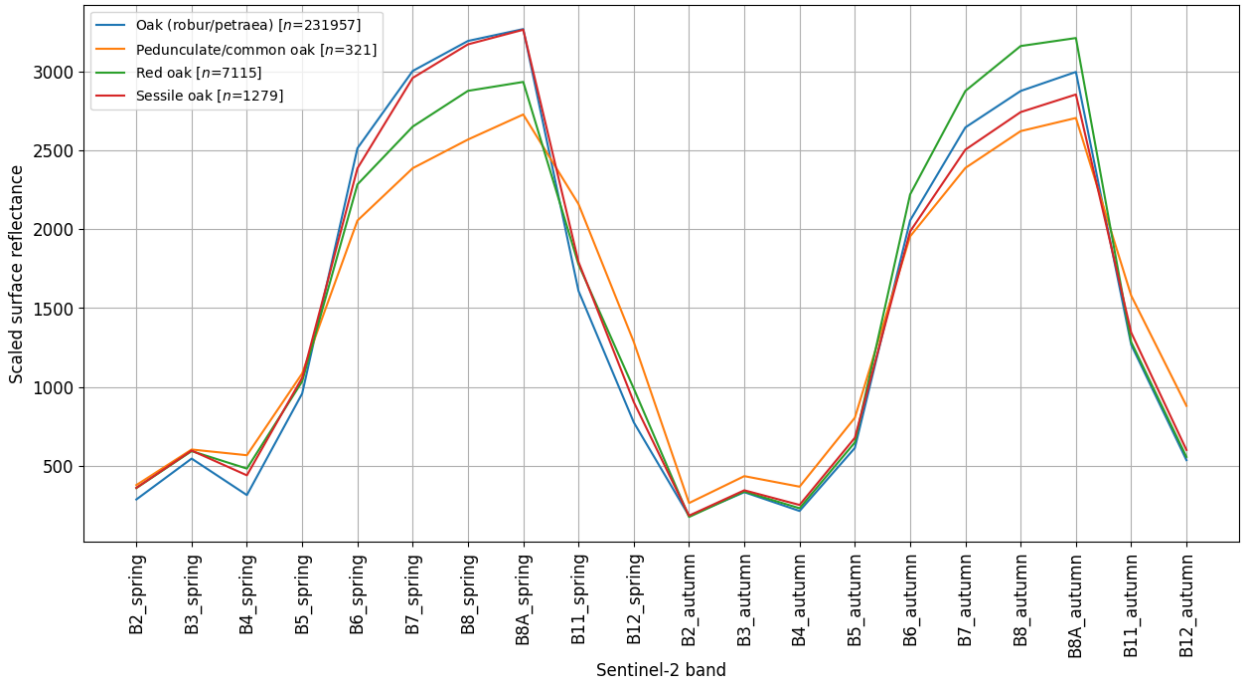

Figure 2: Mean S2 band SR measurements in the spring and autumn of 2019 for several oak species. The number of *n* samples used to compute each mean band SR measurement is shown in the legend.

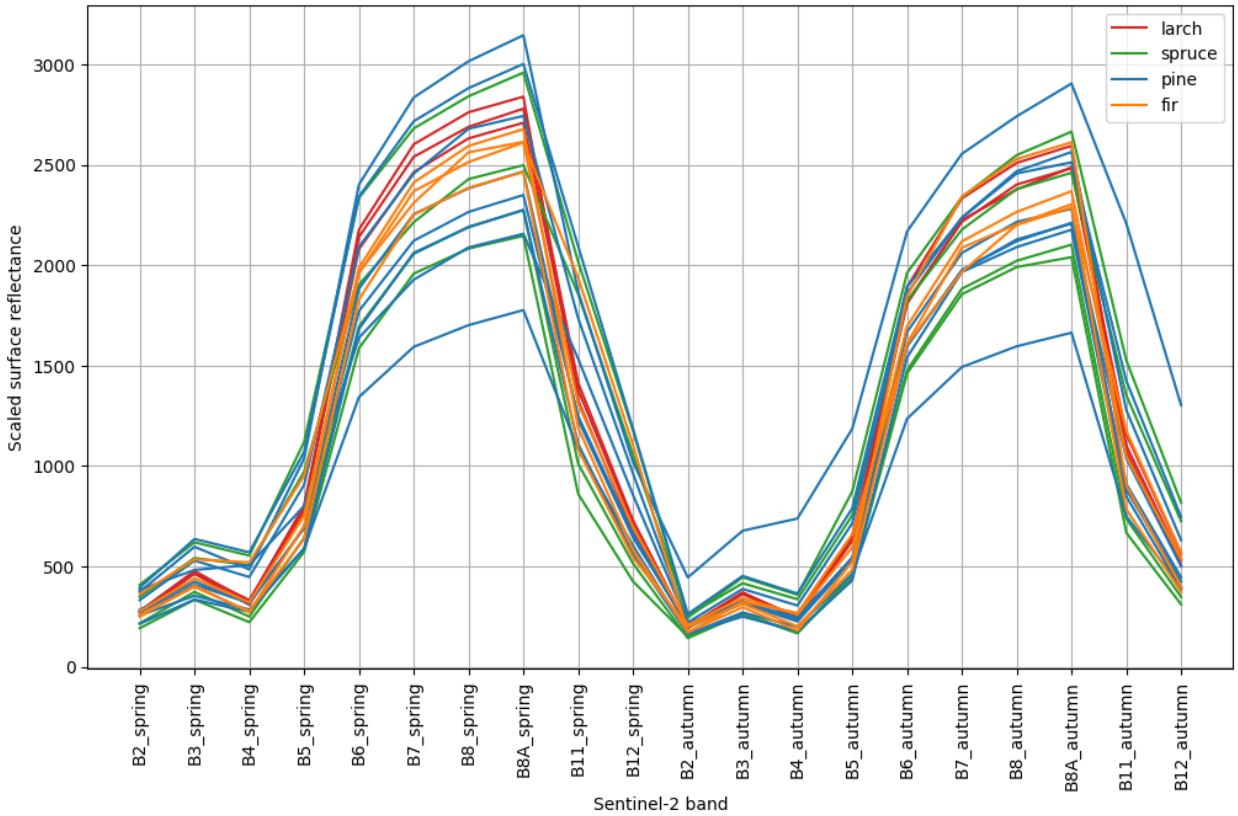

Figure 3: Mean S2 band SR measurements in the spring and autumn of 2019 for larch, spruce, pine and fir species, colour-coded by genus. Note the spectral diversity of species within the same genus.

2.2 Training data

Composite S2 band data from late spring and autumn of 2019, 2020, 2021 and 2022 were sampled at 10m resolution from mono-species stands in England as defined in the SCDB (i.e. stands where "PRIPCTAREA" is equal to 100). For each pixel, there are 10 S2 band measurements per seasonal composite, meaning there are a total of 80 features per pixel (2 seasonal composites for 4 years). After some initial tests, in order to improve the overall classification performance, the decision was made to group minority species with less than 1000 training examples into two engineered classes: 'Other conifer' and 'Other broadleaf'. This gave a total of 35 classes to be learned by the classifier (Figure 4).

A stratified 3:1 train-test split was applied to the data. The test results in Section 4.1 are associated with the 25% of held-out data not seen during the training (right column in Figure 4).

```
(Sitka spruce            1501537
 Corsican pine            673898
 Scots pine               414699
 Beech                    335224
 Oak (robur/petraea)      258211
 Douglas fir              206141
 Japanese larch           145442
 Norway spruce            120498
 Hybrid larch              78001
 Birch (downy/silver)      76951
 Lodgepole pine            69480
 Sweet chestnut            63517
 Western hemlock           37400
 Ash                       20368
 European larch            19133
 Hybrid poplar             11696
 Western red cedar         11093
 Sycamore                  10687
 Red oak                    7639
 Hazel                      6894
 Alder                      6754
 Grand Fir                  6741
 Other broadleaf            4585
 Other conifer              4022
 Lawsons cypress            3437
 Common alder               2870
 Bishop pine                2522
 Yew                        2405
 Monterey pine              2230
 Holly species              1953
 Sessile oak                1937
 Serbian spruce             1565
 Coast redwood              1527
 Hornbeam                   1192
 Norway maple               1138
 Name: Species, dtype: int64,
```

```
Sitka spruce            1126153
Corsican pine            505423
Scots pine               311024
Beech                    251418
Oak (robur/petraea)      193658
Douglas fir              154606
Japanese larch           109081
Norway spruce             90373
Hybrid larch              58501
Birch (downy/silver)      57713
Lodgepole pine            52110
Sweet chestnut            47638
Western hemlock           28050
Ash                       15276
European larch            14350
Hybrid poplar              8772
Western red cedar          8320
Sycamore                   8015
Red oak                    5729
Hazel                      5170
Alder                      5065
Grand Fir                  5056
Other broadleaf            3439
Other conifer              3016
Lawsons cypress            2578
Common alder               2152
Bishop pine                1892
Yew                        1804
Monterey pine              1673
Holly species              1465
Sessile oak                1453
Serbian spruce             1174
Coast redwood              1145
Hornbeam                    894
Norway maple                854
Name: Species, dtype: int64
```

```
Sitka spruce            375384
Corsican pine           168475
Scots pine              103675
Beech                    83806
Oak (robur/petraea)      64553
Douglas fir              51535
Japanese larch           36361
Norway spruce            30125
Hybrid larch             19500
Birch (downy/silver)     19238
Lodgepole pine           17370
Sweet chestnut           15879
Western hemlock           9350
Ash                       5092
European larch            4783
Hybrid poplar             2924
Western red cedar         2773
Sycamore                  2672
Red oak                   1910
Hazel                     1724
Alder                     1689
Grand Fir                 1685
Other broadleaf           1146
Other conifer             1006
Lawsons cypress            859
Common alder               718
Bishop pine                630
Yew                        601
Monterey pine              557
Holly species              488
Sessile oak                484
Serbian spruce             391
Coast redwood              382
Hornbeam                   298
Norway maple               284
Name: Species, dtype: int64
```

Figure 4: Tree species class abundances for England. (Left) All data prior to a stratified 3:1 train-test split, (Middle) Training set, (Right) Test set. Each sample corresponds to a 10x10m^2 region within a mono-species stand. Note the large imbalance in species representation.

3 Model

3.1 XGBoost classifier

As shown in Figure 4, there is a massive class imbalance in the training data, with the majority class (Sitka spruce) outnumbering the minority class (Norway Maple) by a factor of ~1000. This is a common occurrence in real-world machine learning applications. Training a naive classifier will likely result in the model learning it can score good overall accuracy by simply assigning most training examples to the majority class. To mitigate this issue, during the training, a sample weight equal to the inverse of the class frequency was attached to each training sample. This weighting gives more importance to rare classes when they are seen by the model during training. Also, because of cloud and shadow masking during the construction of the S2 seasonal composites, there will be inevitable missing entries for some pixels in some seasons in one or more years. The XGBoost classification algorithm supports weighting samples and can handle sparse feature matrices (among a large number of other attractive properties), and so was the algorithm of choice for this project.

The XGBoost algorithm is an ensemble classifier, which sequentially trains a collection of weaker learners, which collectively make predictions on data. Consequently, a model confidence score for a prediction can be computed as the weighted sum of all tree predictions for a given class. This weighted sum is transformed to a "probability" via a softmax function, such that the probabilities across all classes sum to 1.

Due to the size of the data and consequent processing times, hyperparameter tuning for the 'n_estimators', 'max_depth', 'min_child_weight' and 'learning_rate' was completed via a grid search cross validation on a subset of the training data, with a maximum of just 10,000 samples per class. All other hyperparameters were set to their defaults. The model that was fit to the training data with the best performing hyperparameters is shown in Figure 5.

```
clf=xgb.XGBClassifier(booster='gbtree', n_estimators=500, max_depth=6, min_child_weight=1,
                      learning_rate=0.3,
                      objective='multi:softprob', seed=0)
```

Figure 5: XGBoost classifier hyperparameter settings, as defined in the Python API.

3.2 Discussion of model biases

All models are biased approximations of reality, and it is important to communicate these biases to those making use of any derived products. Below, we outline some of the more important biases users should be aware of.

3.2.1 Training data

The training data consists of mono-species stands, and so assumes that the area covered by a given pixel contains only a single tree species. Blending of spectra from multiple species within a 10x10 m^2 area may produce inaccurate and overly confident predictions. Given this, predictions are expected to be less accurate on NFI stands labelled as Mixed mainly conifer, Mixed mainly broadleaved or Assumed woodland, in addition to low density stands.

Also, most species are not uniformly distributed throughout England e.g. most Sitka spruce is located near the Scottish border, Beech is mostly distributed in the south etc. It is well known that both the intensity and duration of the growing season of species is greater in the south, and this may result in spectral differences within the same species located in different regions. Consequently, the accuracy of predictions for a given class far from the majority of training examples is expected to be worse than the test results would imply.

Lastly, for many species, much of the training data comes from mature stands, and younger trees are likely to differ spectrally to some greater or lesser extent, depending on the difference in age.

3.2.2 Species hypothesis space

We’ve trained the model to discriminate a large number of classes, such that we’re approximating the challenging reality of the tree species classification problem as well as we reasonably can. However, additional simplifications to the model can be made dependent on the use-case, as the importance of the accuracy of the classification of certain species may be problem specific.

3.2.3 Clearfelling and storm damage close to and after 2020

For this work, we have used the most recently available version of the National Forest Inventory (NFI), which was released in 2020, and we have not masked clearfelled stands, or those damaged by storms since this NFI release. We note that for England, a substantial amount of Sitka spruce near the Scottish borders was damaged by Storm Arwen in late 2021 but given the overall representation of this species in the training data, we do not assume this should badly bias the model predictions for this species.

3.2.4 Missing values

An attractive quality of the XGBoost algorithm is its capability to construct decision trees that learn how to best handle missing values during the training. Missing data is however no substitute for the real thing, so we expect predictions to be worse in north-westerly areas where cloud cover is common. Further, the cloud and shadow masking applied to the composites is not perfect, and so the overall quality of composites generated in cloudy areas is likely to be poorer than for the rest of England.

4 Results

4.1 Test results

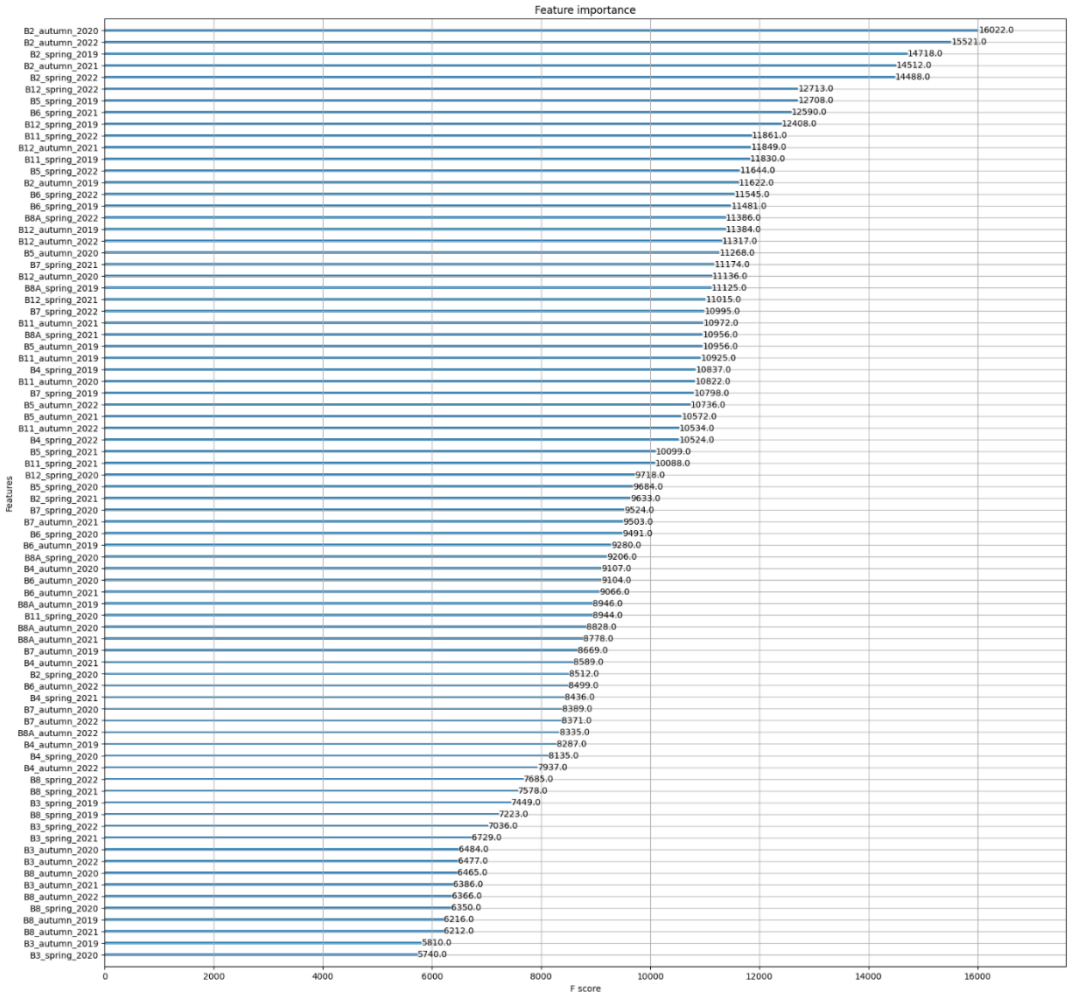

Figure 6: Features ranked by importance, with their associated F scores.

The feature importance is shown in Figure 6. Encouragingly, we see B2 (blue), B5 (vegetation red-edge), B6 (vegetation red-edge), B11 (SWIR) and B12 (SWIR) band information regularly appearing as the most important features, suggesting the model has captured useful information in these bands over multiple years, and there is no single feature that dominates the others. Conversely, B3 (green) and B8 (NIR) band data are consistently ranked as the least important features.

The classification report, (Figure 7) and confusion matrix (Figure 8) show good performance across all classes. Unsurprisingly, classification accuracy appears to be higher for broadleaf species than conifer species in general, with the notable exception of Birch. Due to the weighting scheme used during the training, the overall accuracy of 89% is very similar to the balanced class accuracy of 90%. While encouraging, we stress that these accuracies are from test data that has been sampled from the same stands as the training data (i.e. some pixels from a stand end up in the training set, while others from the same stand may go into the test set), so they are likely optimistic. Further, since this data is from single species stands, these accuracies do not reflect the model's predictive performance on discriminating dominant species within mixed species 10x10m^2 regions (Section 3.2.1).

Classification Report:

	precision	recall	f1-score	support
Alder	0.94	0.92	0.93	1689
Ash	0.88	0.94	0.91	5092
Beech	0.95	0.93	0.94	83806
Birch (downy/silver)	0.58	0.83	0.68	19238
Bishop pine	0.92	0.97	0.95	630
Coast redwood	0.89	0.86	0.88	382
Common alder	0.93	0.94	0.93	718
Corsican pine	0.92	0.86	0.89	168475
Douglas fir	0.82	0.88	0.85	51535
European larch	0.83	0.91	0.87	4783
Grand Fir	0.86	0.89	0.88	1685
Hazel	0.97	0.95	0.96	1724
Holly species	0.97	0.94	0.96	488
Hornbeam	0.94	0.94	0.94	298
Hybrid larch	0.75	0.88	0.81	19500
Hybrid poplar	0.85	0.90	0.88	2924
Japanese larch	0.78	0.87	0.82	36361
Lawsons cypress	0.91	0.86	0.89	859
Lodgepole pine	0.78	0.94	0.85	17370
Monterey pine	0.97	0.94	0.96	557
Norway maple	0.94	0.81	0.87	284
Norway spruce	0.67	0.87	0.75	30125
Oak (robur/petraea)	0.89	0.91	0.90	64553
Other broadleaf	0.89	0.84	0.87	1146
Other conifer	0.87	0.79	0.83	1006
Red oak	0.94	0.93	0.93	1910
Scots pine	0.78	0.80	0.79	103675
Serbian spruce	0.92	0.88	0.90	391
Sessile oak	0.93	0.90	0.91	484
Sitka spruce	0.99	0.92	0.95	375384
Sweet chestnut	0.84	0.94	0.89	15879
Sycamore	0.90	0.92	0.91	2672
Western hemlock	0.77	0.91	0.84	9350
Western red cedar	0.91	0.93	0.92	2773
Yew	0.95	0.96	0.95	601
accuracy			0.89	1028347
macro avg	0.88	0.90	0.88	1028347
weighted avg	0.90	0.89	0.89	1028347

Figure 7: Classification report on the 25% of held-out test data.

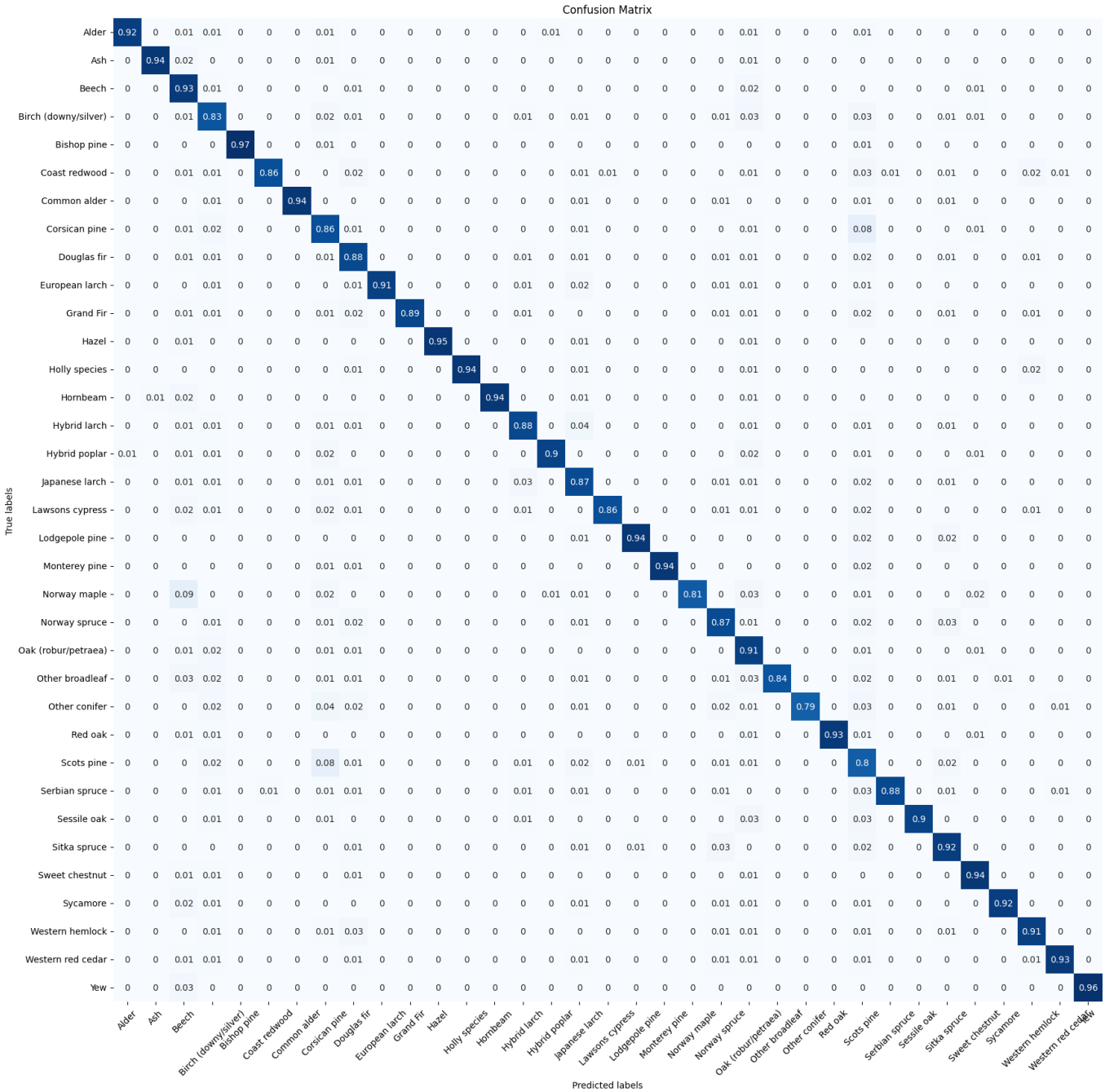

Figure 8: Normalised confusion matrix for the 25% of held-out test data.

4.2 Post-processing and derived products

Predictions are made on the S2 data at the pixel-level but may be further processed depending on the particular needs of downstream tasks. We produce 3 basic derived products:

1) Pixel-level prediction raster for all species. The raster has two bands, the first of which is the integer class label for the most probable species (shown in Table 1), and the second is the model percentage confidence in that prediction.
2) Per-species prediction rasters. This is a single band raster, where the pixel values are the model percentage confidences for regions in which the species under consideration is deemed the most probable. A 3x3 median filter is applied to the predictions to smooth them.
3) Per-species polygons. This is an extension to product 2), where after the median filtering step, the pixels are thresholded to be greater or equal to a user-defined model confidence (default is 60%), before polygonising them with a minimum area criterion (default is 0.1 ha, or 10 connected pixels). The polygon features include an ID number, the area in ha, and the median model prediction confidence for the polygonised pixels.

**** Note: Only product 1 has been released ****

Examples of the three products are shown in Figure 9. The CRS of the products, and the data from which they are derived, is EPSG:32630.

Table 1: Integer class labels for the 35 species classes in the first band of Product 1.

Species	**Integer class label in Product 1**
Alder	0
Ash	1
Beech	2
Birch (downy/silver)	3
Bishop pine	4
Coast Redwood	5
Common alder	6
Corsican pine	7
Douglas fir	8
European larch	9
Grand fir	10

Hazel	11
Holly species	12
Hornbeam	13
Hybrid larch	14
Hybrid poplar	15
Japanese larch	16
Lawsons cypress	17
Lodgepole pine	18
Monterey pine	19
Norway maple	20
Norway spruce	21
Oak (robur/petraea)	22
Other broadleaf	23
Other conifer	24
Red oak	25
Scots pine	26
Serbian spruce	27
Sessile oak	28
Sitka spruce	29
Sweet chestnut	30
Sycamore	31
Western hemlock	32
Western red cedar	33
Yew	34

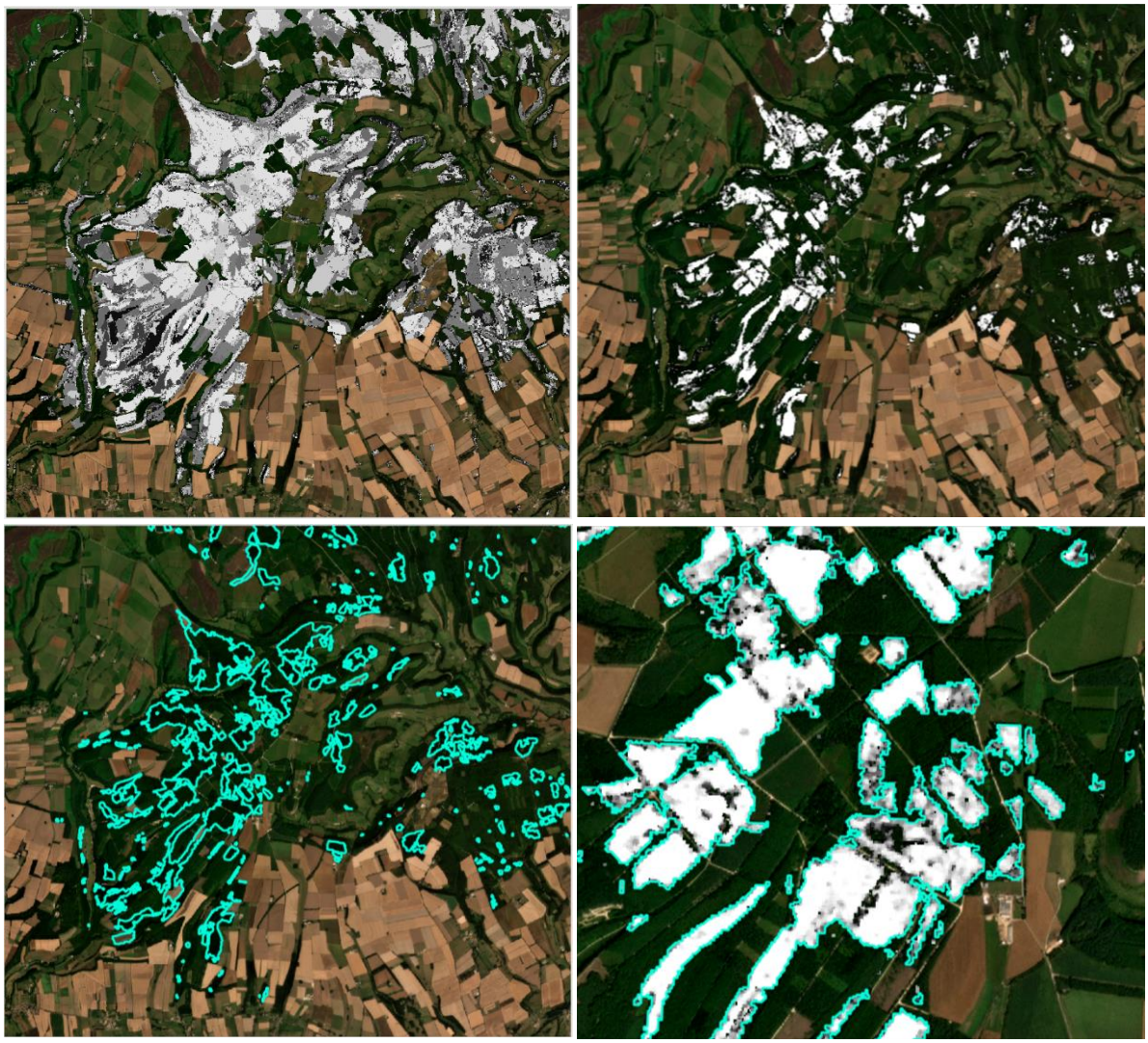

Figure 9: Examples of the available products; the region is North Ridings Forest Park (North York Moors). (*Top left*) The first raster band of Product 1, where each pixel value is an integer corresponding to the integer label of the most probable class. (*Top right*) Product 2 for Sitka spruce, where each pixel is the model percentage confidence in the prediction; the whiter the pixel the more confident the model is. (*Bottom left*) Product 3 for Sitka spruce, where the predictions from Product 2 have been thresholded to >=60% model confidence and polygonised with a minimum area of 0.1 ha. (*Bottom right*) A close-up of Product 3 superimposed on Product 2 for Sitka spruce.

5 Validation

5.1 Validation by Forestry Commission

Field work was undertaken by the Plant Health team within FC to validate the accuracy of Norway spruce predictions within the *Ips typographus* DMA in southeast England. This validation data was passed over to FR in March 2024, and the results are summarised below.

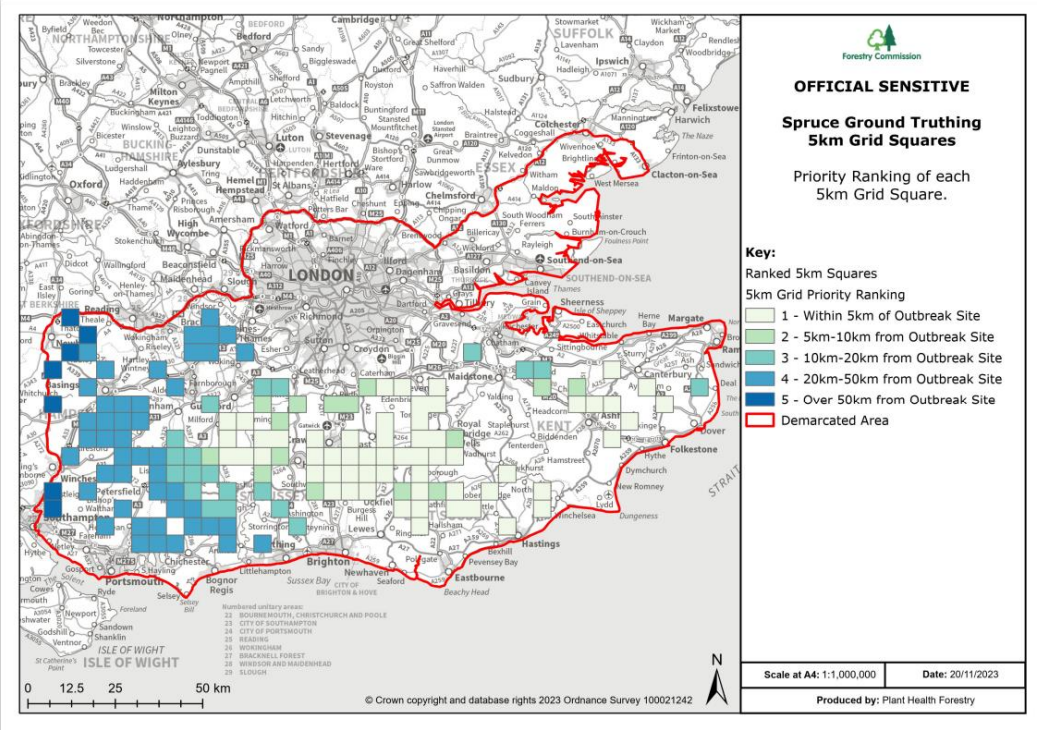

Figure 10: Within the DMA, 5km grid squares were defined and ranked by priority, measured by distance from known outbreak sites.

590 prediction polygons within the DMA (Figure 10) were visited for ground validation. These polygons do not intersect with regions in which species information is already known e.g. from the SCDB. For 567 of these polygons, the dominant species was recorded. Norway spruce was found to be dominant for 390 of these sites, giving a precision (TPs / TPs + FPs) of 69%. Encouragingly, this broadly agrees with the precision as measured on our held-out test data of 67% for this particular species (Figure 7).

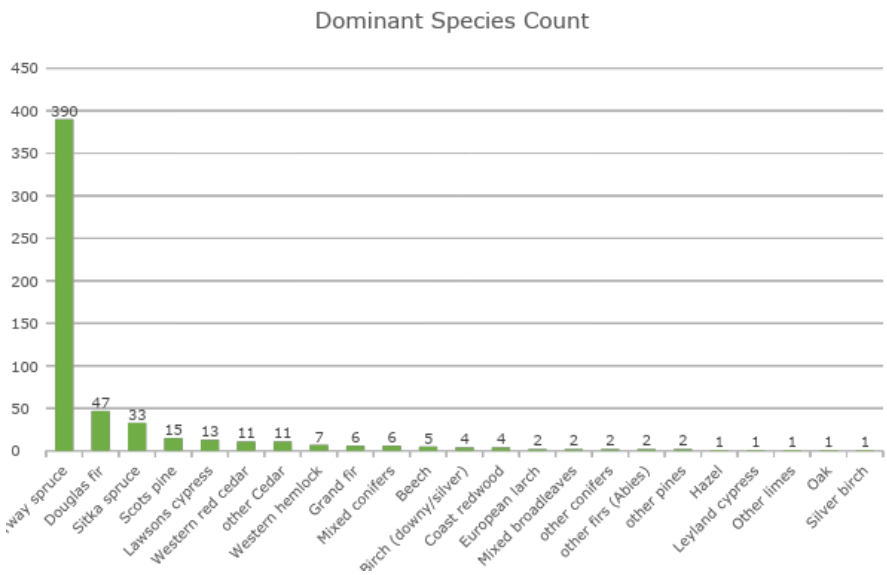

Figure 11: Bar chart of the dominant species for 567 polygons predicted as Norway spruce.

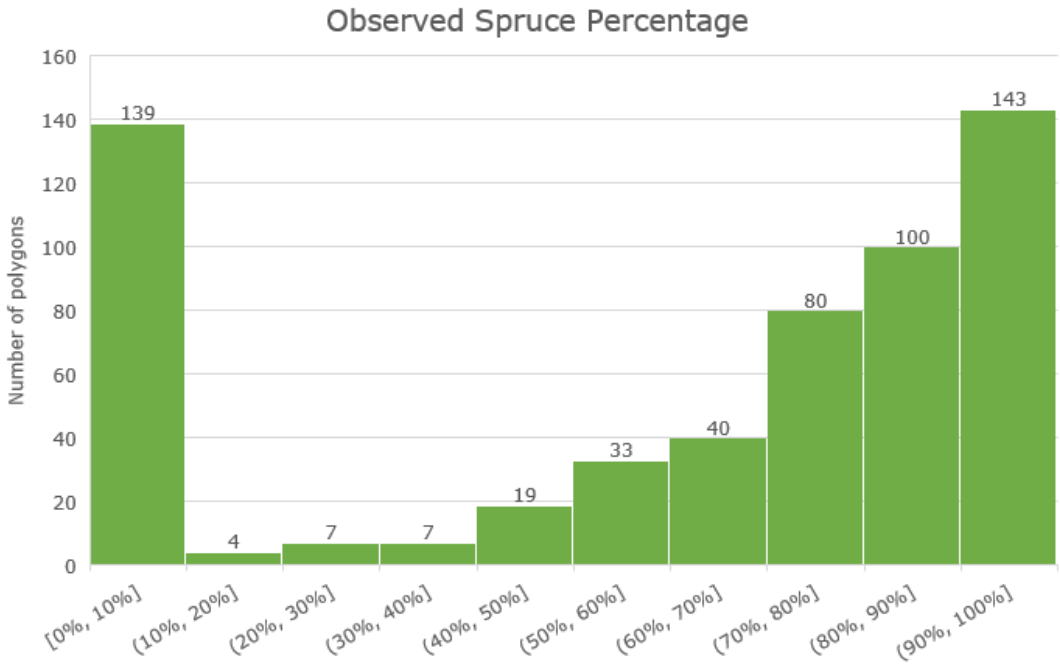

Figure 12: Bar chart showing the percentage abundance of spruce within 572 of the predicted Norway spruce polygons.

572 of polygons had recorded information on the % abundance of spruce within the predicted area (Figure 12). As expected, most polygons contained 90 - 100% spruce, and 25% of all prediction polygons had at least 90% spruce. This makes sense, as these sites will be most similar to the mono-species SCDB training data. 139 of these polygons contained 0 - 10% spruce, and are likely false positives associated with Douglas fir, Sitka spruce or Scots pine (Figure 11). It has been found that these species are phenologically and spectrally similar to Norway spruce, so this result is not surprising, and indicates a limit of the optical Sentinel-2 data used to disentangle these species.

Figure 13: Norway spruce predictions (green) and on-ground observed Norway spruce extent (purple).

Lastly, it was generally found that the Norway spruce predictions underestimated the true extent of spruce within a region. This again makes sense, being a purposeful design choice in both the training of the model and thresholding during the construction of the map. Consequently, we expect the false negative rate to exceed the false positive rate for most species, although there are challenges with designing surveys to accurately measure the former.

5.2 Follow up comments on the validation by FR

Data exploration for the incorrectly predicted sites suggested that there was nothing *obviously* bad with the spectra e.g. not seeing lots of missing data due to clouds,

or sites that have been clearfelled well before 2022 (the Sentinel-2 data used spans 2019 - 2022 inclusive). The upside of this is that there's nothing terribly bad going on with the data, but the flip side is that this means there likely isn't a straightforward solution to improve the accuracy (at least for Norway spruce predictions within the DMA). The solution is to generate more discriminating features (we don't think we can easily do better with Sentinel-2 alone), but no data set immediately springs to mind. Naively, one would assume the scattering features measurable with Sentinel-1, or structure metrics computed from Lidar etc. would also be similar for these species.

It was also found that there also doesn't seem to be a particularly strong connection between the accuracy of the predictions and their distance from the monocultural SCDB training data samples near or within the DMA. This is good news, suggesting that the model generalises reasonably well.

It also seems Norway spruce was incorrectly predicted over cedar sites. This also makes some sense, as only one cedar species (Western red cedar) was represented in the training data, and there weren’t many training examples. This is perhaps a simpler fix, where we designate cedars as a new distinct class for the model to learn, but would need to do a bit of data exploration to see if it's justifiable to clump multiple cedar species together like this. This does support our decision to try and get the model to disentangle the true complexities of species mapping by defining many classes.

5.3 Validation with NFI squares

Validation was also completed for several other species with reference to historical field data obtained within NFI squares. Due to not having the exact locations of species within any given square, in order for this analysis to be meaningful, cuts were made to compare only the model predictions intersecting with squares containing a single species, and with a representation of more than 2500 ha within the square. This is therefore a crude comparison due to the lack of geolocated validation data. Nonetheless, a confusion matrix for the predicted labels within the filtered squares is shown in Figure 14 below. This suggests broadly good performance across species and is consistent with prior results on held-out test data from mono-species SCDB stands e.g. confusion between Japanese and Hybrid larch, or Norway spruce with Sitka spruce and Scots pine. Further, Figure 14 also suggests that the model is likely over-predicting Oak in regions dominated by species either poorly represented in the training data, or not at all. This likely arises from the class imbalances in the data, in addition to spectral similarities.

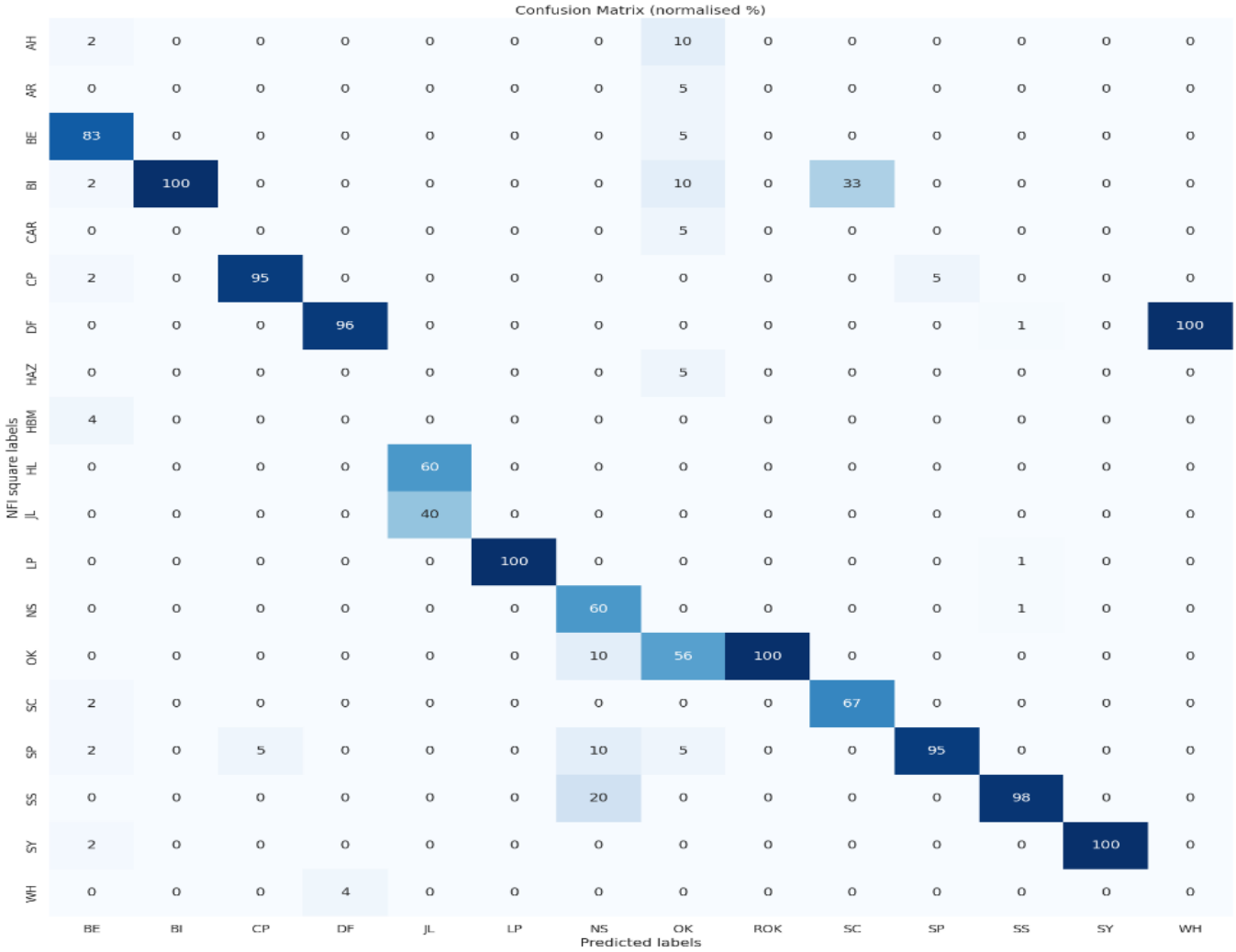

Figure 14: Normalised confusion matrix for predicted labels intersecting with mono-species NFI squares.

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